

Oscar Andrés Flores Gaitán

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Education

Ph.D. in Physics

Department of Physics & Astronomy, University of Delaware

Newark, DE

Aug 2026 –

- Starting Fall 2026.

Honors B.S. in Physics

Universidad del Valle de Guatemala

Guatemala City

Jan 2022 – May 2026

- Awaiting thesis defense.
- Thesis: The Planetary Architecture of Barnard's Star: A Bayesian Approach with Multi-Instrument Radial Velocity Data

Research Interests

- Exoplanet detection and characterization, planet formation, and evolution.
- Application of machine learning and advanced statistical methods to astronomical time-series and spectral data.

Publications

2. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V. "*How many terrestrial planets orbit Barnard's Star?*". [Manuscript in preparation].
1. Zhao, L.L. et al., **including Flores Gaitán, O.A.**, "*The Extreme Stellar-Signals Project IV*". [Manuscript in preparation].

Talks & Posters

6. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., Rubio-Herrera, E., "*How many terrestrial planets orbit Barnard's Star?*", **Contributed Talk** at APS Mid-Atlantic Section Meeting 2025, Penn State Harrisburg, Middletown, PA, November 15, 2025. [Link](#) [🔗](#)
5. **Flores Gaitán, O.A.**, "*Hunting for Exoplanets: Real Planets or Stellar Processes?*", **Invited Talk** at Universidad del Valle de Guatemala's Physics Week, Guatemala City, October 25, 2025.
4. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., "*Disentangling Planetary and Stellar Signals in Barnard's Star*", **Poster** at Physics Science Fair at Universidad del Valle de Guatemala's Physics Week, Guatemala City, October 24, 2025.
3. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., "*Disentangling Planetary and Stellar Signals in Barnard's Star*", **Virtual Poster** at NASA's Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology, Johns Hopkins University, Washington, DC, July 28-31, 2025.
2. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., "*Disentangling Planetary and Stellar Signals in Barnard's Star*", **Virtual Poster** at Sagan Summer Workshop, NASA Exoplanet Science Institute, California Institute of Technology, Pasadena, CA, July 23, 2025. [Link](#) [🔗](#)
1. **Flores Gaitán, O.A.**, "*Detecting and Characterizing Exoplanets Around M-Dwarfs with TESS*", **Invited Talk** at Universidad del Valle de Guatemala's Astronomy Club, Guatemala City, May 9, 2025.

Research Experience

Undergraduate Visiting Researcher

Department of Physics & Astronomy, University of Delaware

Advisor: Prof. Sarah Dodson-Robinson.

Co-advisor: Victor Ramirez Delgado

Newark, DE

Jun 2025 – present

- Evaluated advanced statistical methods to distinguish genuine exoplanetary signals from stellar noise in high-precision radial velocity (RV) data.
- Implemented Gaussian Process (GP) regression and frequency-domain analysis to mitigate stellar variability and improve the reliability of RV detections.
- Analyzed multi-instrument RV datasets to determine the planetary architecture of Barnard's Star, utilizing advanced statistical metrics for model validation.

Undergraduate Researcher

Universidad del Valle de Guatemala

Advisor: Prof. Eduardo Rubio-Herrera

Guatemala City

Nov 2024 – May 2025

- Processed and analyzed photometric time-series data from the TESS mission to identify exoplanet candidates.
- Developed an algorithm to detect transits and extract orbital parameters from stellar light curves.

Academic Appointments & Teaching

Teaching Assistant

Department of Physics & Astronomy, University of Delaware

- Starting Fall 2026.

Newark, DE

Aug 2026 –

Teaching Assistant & Laboratory Assistant

Physics Department, Universidad del Valle de Guatemala

- Designed and implemented laboratory guides for undergraduate physics courses, assisted in the setup and supervision of laboratory sessions, and graded reports.
- Served as Teaching Assistant for 7 undergraduate physics courses, including Mechanics I and Physics I-III.

Guatemala City

Jan 2024 – May 2025

Tutor

Universidad del Valle de Guatemala

- Provided 300+ hours of one-on-one tutoring for over 30 high school and college students, prepared personalized study materials, and tracked student progress.
- Precalculus, Calculus I-III, Calculus for Business I-II, Differential Equations, Physics I-III

Guatemala City

Jun 2022 – present

Certifications

Professional Certificate in Data Science and Machine Learning

HarvardX (Harvard University)

- Introduction to Data Science with Python
- Machine Learning and AI with Python

In progress

Data Analysis with Python Certification

FreeCodeCamp

- Python for Everybody
- Data Analysis with Python

Completed

Workshops & Conferences

Alpha-Cen Astronomy Summer School

Alpha-Cen (Central American and Caribbean Astrophysics)

Remote

Aug 2025

Workshop on Astrophysics of High Energies and Cosmology

Universidad Nacional Autónoma de México (UNAM)

Guatemala City

Nov 2024

Awards & Achievements

- Selected for Summer Visiting Researcher Program at the Department of Physics and Astronomy, University of Delaware, out of 147 applicants.
- Top 10 National Mathematics Olympiad 2020
- Achieved university's most accurate experimental measurement of gravitational acceleration (**g**)
- 3x Distinguished Student Award, Universidad del Valle de Guatemala
- IRONMAN 70.3 Finisher and Federated Triathlete

Technical Skills

Programming Languages: Python, L^AT_EX, R (Intermediate)

Data Analysis and Probabilistic Modeling: Statistical inference, Gaussian Processes (GPs), Markov Chain Monte Carlo (MCMC), Bayesian modeling, time-series analysis

Machine Learning: Scikit-learn, XGBoost, Random Forest, K-Nearest Neighbors (KNN), Logistic Regression

Astronomy Tools: Astroquery, ExoCTK, Lightkurve, AstroPy, NWAELCH, PyMC

Languages: Spanish (Native), English (C1), German (C1)

Outreach & Leadership

- Created and published educational physics content on YouTube, including a laboratory guide video with over **1,100 views**. [Link](#) 
- Authored a physics laboratory guide widely adopted by students and instructors at Universidad del Valle de Guatemala.
- Founded and managed a popular educational Instagram account focused on geography and linguistics — reached over **15,000 followers** and **5 million total views**. [Link](#) 
- Member, Physics Students Association — organized science talks and led outreach initiatives for physics students.
- Lead Organizer, Physics Week 2025 — Led a multi-day event featuring expert talks and workshops on diverse topics in physics.
- Workshop Developer & Instructor — Designed and taught the workshop "How to Write an Academic CV", providing foundational career guidance for first-year physics students. October 2025.
- Teaching Assistant — Collaborated on a specific lecture regarding Extragalactic Astronomy, helping translate high-level physics for a high school audience. Astro Antigua. December 2025.
- Workshop Developer & Instructor — Designed and taught the workshop "How to apply to internships in the US", providing foundational career guidance for undergraduate physics students in Latin America. February 2026.